



고려대학교 약학대학
KOREA UNIVERSITY COLLEGE OF PHARMACY

KU-The Future
110 years and beyond

Microbiology



Fall semester, 2026

❖ Classification of bacteria

► Gram-positive bacteria

- phylum Bacillota (previously, Firmicutes)
 - classes : Bacilli, Clostridia, Negativicutes, etc.
 - class **Clostridia**
 - highly polyphyletic: having various shapes and sizes
 - all members: obligate anaerobes / endospores
 - genera: *Clostridium*, etc.
 - genus *Clostridium*
 - the largest genus (~100 species) in this class
 - most species are saprophytic organisms
 - five main species responsible for disease in humans
 - : *C. botulinum*, *C. tetani*, *C. difficile*, *C. perfringens*, *C. sordellii*



C. tetani

❖ Classification of bacteria

▶ Gram-positive bacteria / phylum Bacillota

· Class Clostridia / Genus *Clostridium*

– *C. botulinum* :

- etymology: L. botulus (“sausage”) / mainly found in soil and water / zoonosis (?); infecting animals, but not zoonotic (having zoonotic relevance)
- producing botulinum toxin (보툴리눔 독소) in food or wound
: protein produced under anaerobic conditions
→ causing botulism (called botulinus intoxication; 보툴리눔 식중독, 보툴리누스증)
- three ways for a disease to develop:
 - by ingestion of toxin from foods (foodborne botulism)
 - by colonization of the digestive tract by bacteria in children (infant botulism)
 - by contamination of a wound by bacteria (wound botulism)
 - person-to-person / direct animal-to-person transmission: not occur
- symptoms: paralysis by muscle weakness
: obstacle to eye movements, chewing, swallowing, talking, respiration, etc.
→ highly fatal (70% → 5–10% of lethality)
- diagnosis: detection of botulinum toxin in clinical specimens



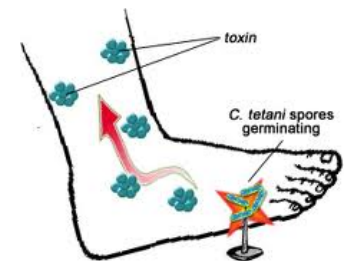
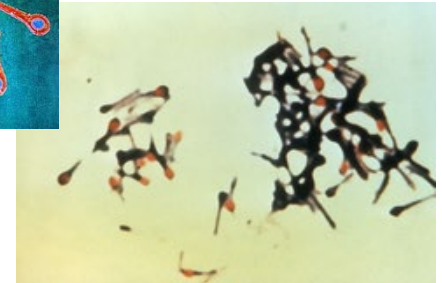
❖ Classification of bacteria

▶ Gram-positive bacteria / phylum Bacillota

· Class Clostridia / Genus *Clostridium*

– *C. tetani*:

- be everywhere around us; mainly in soil (and GI tract of animals) / infecting animals, but not zoonotic
- tetanus (파상풍)
 - etymology: G. tetanos (“taut”) and G. teinein (“to stretch”)
 - caused by the infection through wound contamination
 - : often involves a cut or deep puncture wound
 - caused by tetanospasmin (파상균 강직 독소)
 - : a neurotoxin produced by this bacteria / acting at several sites within the CNS
- symptoms:
 - unopposed muscle contraction / painful muscular spasms
 - rigid smile, lock-jaw, rigid and arched back, seizure, etc.
 - respiratory failure
 - lethality: up to 40% of cases (lethality in children: ~90%)
- vaccination: DPT vaccine (diphtheria, tetanus, pertussis)



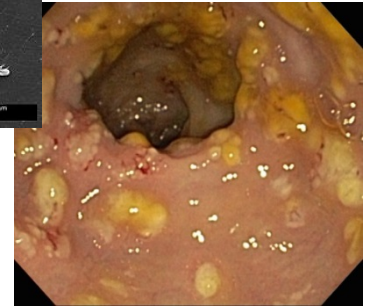
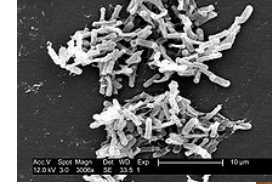
❖ Classification of bacteria

▶ Gram-positive bacteria / phylum Bacillota

· Class Clostridia / Genus *Clostridium*

– *C. difficile* (*Clostridiodes difficile*):

- can be a minor normal component of colonic flora in both humans and animals / zoonosis (?); infecting animals, but not zoonotic (having zoonotic potential)
- can flourish when other bacteria (normal flora) in the gut are killed during antibiotic therapy (e.g. clindamycin)
- producing two toxins: toxin A (enterotoxin; 장독소) and B (cytotoxin; 세포독소)
- causing antibiotic-associated pseudomembranous colitis (위막성 대장염)
 - severe inflammation of the colon
 - offensive-smelling diarrhea, fever, abdominal pain, and (in severe cases) life-threatening complications (such as toxic megacolon)
 - toxic megacolon (중독성거대결장증) : an acute form of colonic distension
- nosocomial infection



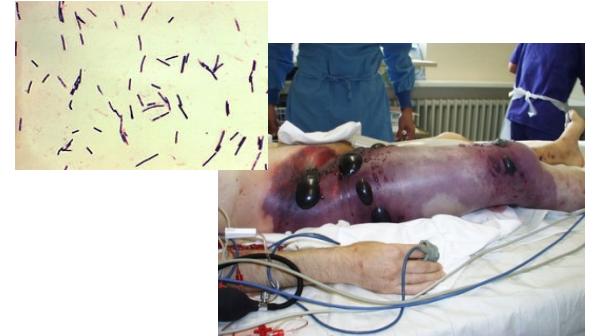
❖ Classification of bacteria

▶ Gram-positive bacteria / phylum Bacillota

· Class Clostridia / Genus *Clostridium*

– *C. perfringens* (formerly, called *C. welchii*)

- ever present in nature and intestinal canal of humans and animals (a component of the normal flora) / infecting animals, but not zoonotic (having zoonotic relevance)
- causing diseases by toxins produced by the bacteria
 - alpha, beta, epsilon, iota, and enterotoxin (CPE; 장독소)
 - from food poisoning to gas gangrene
 - food poisoning: **enterotoxin (CPE)**; one of the most common causes of food poisoning / abdominal cramping, diarrhea, vomiting, etc. → subclinical
 - clostridial necrotizing enteritis (called pigbel; 괴사성 장염): **beta-toxin** / severe than those of food poisoning (50% of lethality)
 - clostridial myonecrosis (근괴사 / gas gangrene; 가스괴저): **alpha toxin** / deadly form of gangrene (producing gas in tissues; necrotic damage)
 - sepsis



❖ Classification of bacteria

► Gram-positive bacteria / phylum Bacillota

• Class Clostridia / Genus *Clostridium*

– *C. sordellii*:

- causing a fatal infection in exceptionally rare cases
 - gynecologic infections in women: reported in post-partum females, medically-induced abortions, injection drug users, trauma cases, etc.
 - infection of the umbilical stump in newborns
 - less than one case per year has been reported since 2000
- pneumonia, endocarditis, arthritis, peritonitis, and myonecrosis
- bacteremia and sepsis



❖ Classification of bacteria

▶ Gram-positive bacteria / phylum Bacillota

- class Negativicutes (previously, Clostridia) > order Veillonellales > family Veillonellaceae > genus *Veillonella*
 - Genus *Veillonella*
 - ex) *V. parvula*
 - normal flora in the oral mucosa of mammals, including humans (but not zoonotic)
 - having fermenting ability of lactate provided by other oral flora (not feeding on other carbohydrates)
 - lactate → acetate + 2 propionate + CO₂ + H₂O
 - cause of dental caries and calculus (called tartar, scale or plaque) and periodontitis
 - rarely cause osteomyelitis and endocarditis

❖ Classification of bacteria

▶ Gram-positive bacteria / phylum Mycoplasmatota

· Class Mollicutes

- previously, classified into phylum Firmicutes
- etymology: L. mollis (“soft”) and L. cutis (“skin”)
 - no cell wall / pleomorphism (vary in form; 다형태성)
 - not affected by β -lactams
- most of the members move through gliding (활주운동)
(but some of them move by twisting)
- the smallest bacteria: physical and genomic size
→ parasites of various animals and plants
- several orders: Acholeplasmatales, Anaeroplasmatales, Entomoplasmatales, Haloplasmatales, Mycoplasmatales

❖ Classification of bacteria

▶ Gram-positive bacteria / phylum Mycoplasmatota

- class Mollicutes

- order **Mycoplasmatales** (> family Mycoplasmataceae)

- : two representative genera: *Mycoplasma* and *Ureaplasma*

- Genus *Mycoplasma* :

- *M. pneumoniae* :

- causing atypical pneumonia (이형폐렴; “walking pneumonia”)
: together with *Legionella pneumophila*, *Coxiella burnetii*, and *Chlamydophila pneumoniae*

- (vs) typical pneumonia: caused by *S. pneumoniae*,
S. aureus, *H. influenzae*, *K. pneumoniae*, *P. aeruginosa*, etc.

- *M. genitalium* : STD / non-gonococcal/specific urethritis (NGU/NSU;
비임균성/특이성 요도염) / cervicitis, pelvic inflammatory disease (PID), etc.

- *M. hominis* : normal flora vs. STD (?) / bacterial vaginosis, PID,
postpartum (endo)metritis / nephropylitis, etc.

❖ Classification of bacteria

▶ Gram-positive bacteria / phylum Mycoplasmatota

- Class Mollicutes

- Order Mycoplasmatales

- : two representative genera: *Mycoplasma* and *Ureaplasma*

- Genus *Ureaplasma* (*U. urealyticum*)

- normal genital flora of both men and women

- : found in about 70% of sexually active humans

- causing NGU, nephropylitis / infertility, stillbirth, premature birth / perinatal (neonatal) pneumonia, bronchopulmonary dysplasia and meningitis, etc.

- normal flora vs. STD (?)

❖ Classification of Bacteria (eubacteria)

▶ Gram-positive bacteria

- **phylum Actinomycetota** (previously, Actinobacteria)
 - one of the most common life form in soil and water
 - play an important role in decomposition of organic materials
 - necessary in organic matter turnover and carbon cycle
 - secondary metabolite producers having high pharmacological and commercial interest (e.g. antibiotics)
 - some of the members inhabit plants and animals, including a few pathogens in humans
 - some of the members form the fungal-like branched networks of hyphae (mycelia) (e.g. genus *Actinomyces*)
 - producing asexual spores, but not endospores
 - formerly, classified to fungi
 - now, done to bacteria: having structures and components of bacterial cell walls and 16S rRNA

❖ Classification of bacteria

▶ Gram-positive bacteria / phylum Actinomycetota

- classes Actinomycetia, etc.
- class **Actinomycetia**
 - orders Actinomycetales, Mycobacteriales, Propionibacteriales, Micromonosporales, Streptomycesetales, Bifidobacteriales, etc.
 - previously taxonomy: orders Actinomycetales, Bifidobacteriales, etc.
 - Actinomycetales and its suborders: Actinomycetaceae, Corynebacteriaceae, Micromonosporaceae, Propionibacteriaceae, Streptomycesaceae, etc.
 - order Bifidobacteriales
 - representative genera:
 - genus *Actinomyces* : Actinomycetales > Actinomycetaceae
 - genus *Corynebacterium* : Mycobacteriales > Corynebacteriaceae
 - genus *Mycobacterium* : Mycobacteriales > Mycobacteriaceae
 - genus *Nocardia* : Mycobacteriales > Nocardiaceae
 - genus *Propionibacterium* : Propionibacteriales > Propionibacteriaceae
 - genus *Micromonospora* : Micromonosporales > Micromonosporaceae
 - genus *Streptomyces* : Streptomycesetales > Streptomycesaceae
 - genus *Bifidobacteria* : Bifidobacteriales > Bifidobacteriaceae

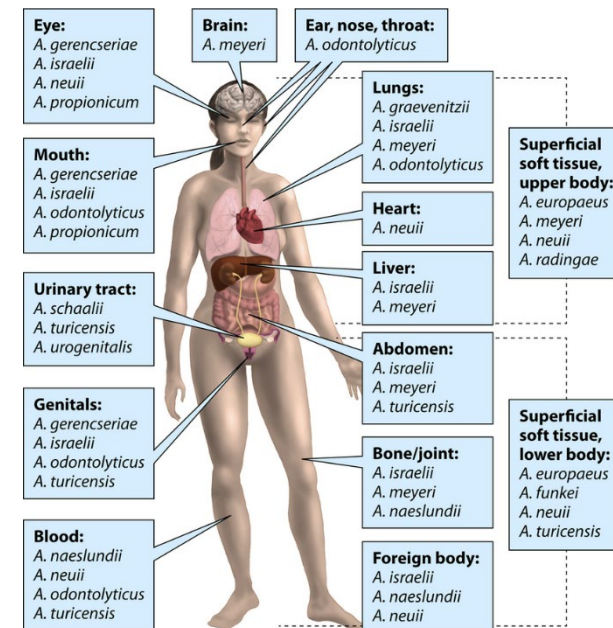
❖ Classification of bacteria

▶ Gram-positive bacteria / phylum Actinomycetota / class Actinomycetia

· order Actinomycetales > family Actinomycetaceae

– genus *Actinomyces*:

- play an important role in soil ecology
: producing many enzymes that help degrade organic plant materials (formation of compost)
- normally present in the gingival area
: commensal flora
- opportunistic infections (actinomycosis) to humans and other mammals, particularly in the oral cavity
- actinomycosis (방선균증): ex) *A. israelii* / the formation of abscesses in the mouth, lungs, the gastrointestinal tract, etc.



❖ Classification of bacteria

▶ Gram-positive bacteria / phylum Actinomycetota / class Actinomycetia

· order Mycobacteriales > family Corynebacteriaceae

– genus *Corynebacterium* :

· non-pathogenic: *C. glutamicum*

– useful in industrial setting: used for producing glutamate and lysine (components of human food, animal feed, and pharmaceutical products)

· pathogenic: *C. diphtheriae* – diphtheria (디프테리아증)

– caused by diphtheria toxin (an exotoxin)

– a contagious disease; upper respiratory tract illness

– transmitted by direct contact (or aerosols?)

– symptoms:

· mild: restricted to the skin

· severe: sore throat, fever, tonsillitis, pharyngitis, myocarditis (about 20% of cases), peripheral neuropathy (about 10% of cases), etc.

– vaccine: DTaP vaccine



Club-shaped



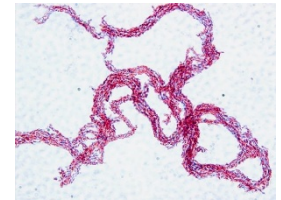
❖ Classification of bacteria

▶ Gram-positive bacteria / phylum Actinomycetota / class Actinomycetia

· order Mycobacteriales > family Mycobacteriaceae

– genus *Mycobacterium* :

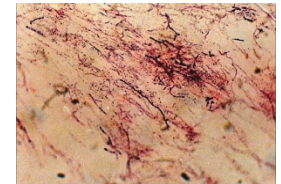
- causing serious diseases in mammals, including humans
- *M. tuberculosis* : tuberculosis (결핵)
- *M. bovis* : bovine tuberculosis (우결핵) / zoonotic
- *M. leprae* : leprosy (Hansen's disease; 한센병, 나병) / zoonotic (?)



· order Mycobacteriales > family Nocardiaceae

– genus *Nocardia* :

- found worldwide in soil and in the oral cavity (especially, healthy gingiva and periodontal pockets) of humans (oral normal flora)
- opportunistic infections: nocardiosis (*N. asteroides*) – slowly progressive pneumonia (pulmonary nocardiosis; cough, fever, dyspnea, etc.) / systemic nocardiosis



❖ Classification of bacteria

▶ Gram-positive bacteria / phylum Actinomycetota / class Actinomycetia

· order Propionibacteriales > family Propionibacteriaceae

– genus *Propionibacterium* (일부가 genus *Cutibacterium* 변경):

· generating propionic acid

– used in the production of propionic acid and cheese industries
/ used as probiotics

– propionic acid:

· inhibits the growth of mold and some bacteria: used as a preservative for human food and animal feed

· has fruit-like odors: used as solvents or artificial flavorings

· commensals of humans and animals

– the skin flora (especially, sweat and sebaceous glands)

– *P. acne* (*Cutibacterium acne*): causing acne

❖ Classification of bacteria

▶ Gram-positive bacteria / division Actinobacteria / order Actinomycetales

- order Micromonosporales > family Micromonosporaceae

- genus *Micromonospora* :

- producing antibiotics: aminoglycosides (e.g. gentamicin, sisomicin, etc.)

- order Streptomycetales > family Streptomycetaceae

- genus *Streptomyces* :

- producing antibiotics: aminoglycoside (e.g. streptomycin, kanamycin, neomycin, etc.)

❖ Classification of Bacteria (eubacteria)

► Gram-positive bacteria / phylum Actinomycetota / class Actinomycetia

· order Bifidobacteriales > family Bifidobacteriaceae

– genus *Bifidobacterium* :

· *Bifidobacterium bifidum* : before the 1960s, be referred to as “*Lactobacillus bifidus*”

· similar features as *Lactobacillus* and *Lactococcus*

– found in gastrointestinal tract, vagina, and mouth of mammals and other animals (symbiotic inhabitants)

: one of the major genera of bacteria making up the colon flora in mammals

– beneficial health effects: regulation of intestinal microbial homeostasis, inhibition of harmful bacteria the colonize and infect the gut mucosa, modulation of immune responses (e.g. reduction of allergic responses), anti-cancer activities, etc.